Measurement of Cryogenic Pellet Ablation Rate and Multi-Species Mass

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Motivation

- Cryogenic Pellets At Columbia (PAC) test stand has been assembled and operated
- Pellets are injected and ablate into plasmas for disruption mitigation systems (DMS), core fueling, density control, and edge-localized mode pacing
- Previous ablation data, primarily in situ from tokamaks, limited by diagnostics and plasma dynamics
- System employs 30 keV electron beam as ablation source
- To-be operated with H_2, D_2, Ne, Ar, and mixed pellets

System Characterization

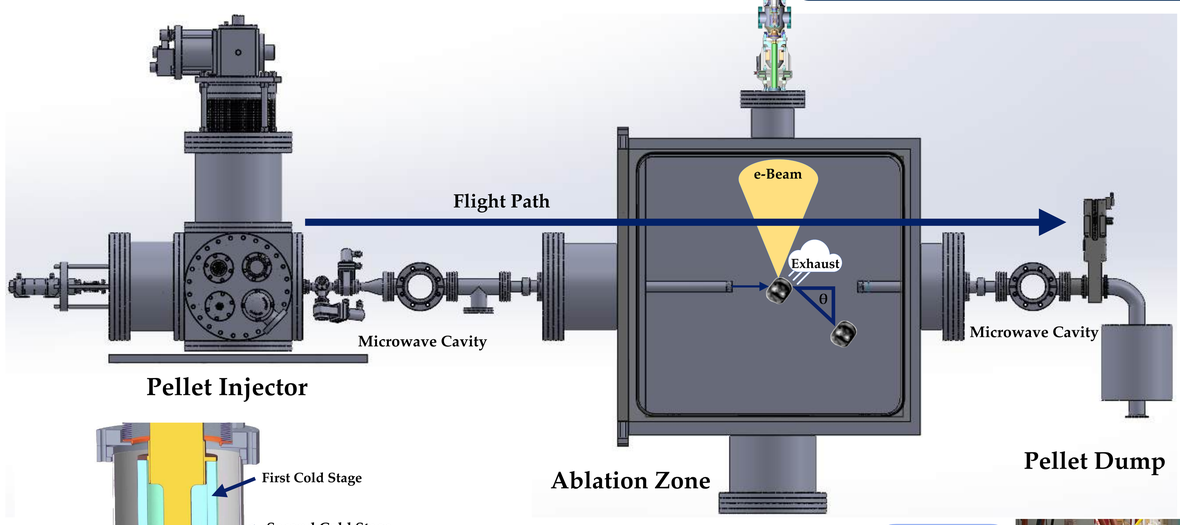
- Initial 6-shot microwave cavity mass measurement

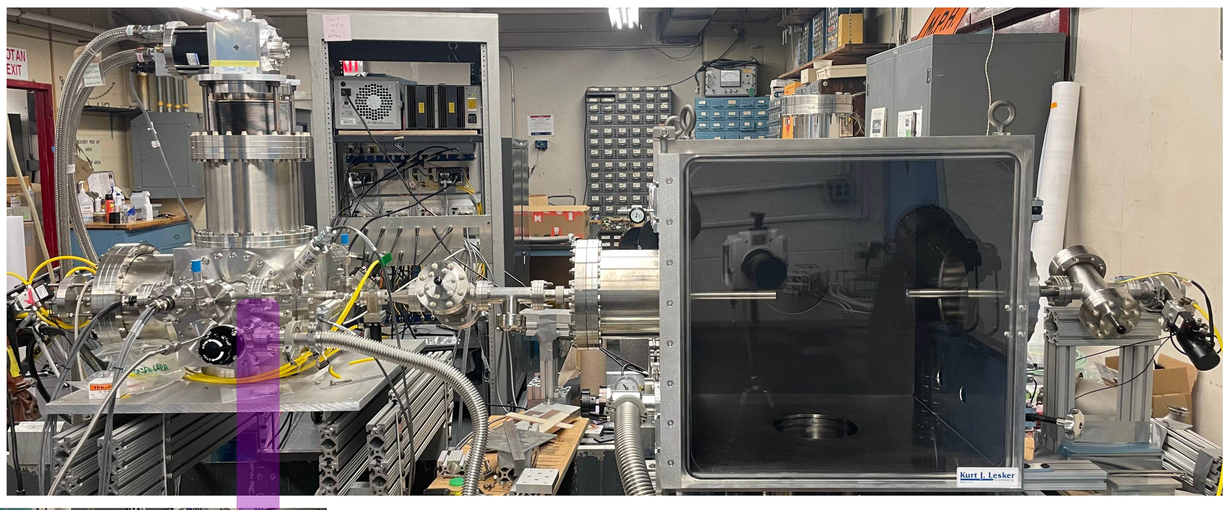
Species	Ideal Mass (mg)	Average Cavity 1 Mass (mg)	Average Cavity 2 Mass (mg)
H_2	41.6 ± 0.2	41 ± 4	41 ± 4
Ne	702 ± 2	630 ± 50	580 ± 50

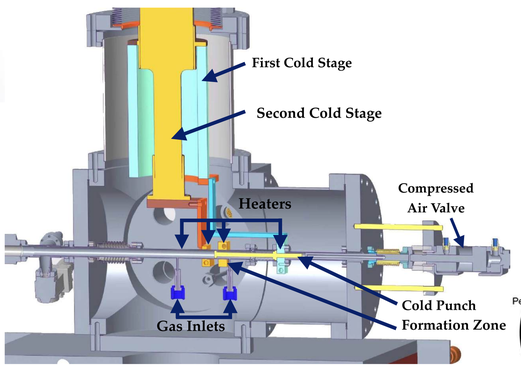
- Initial 6-shot fast camera pellet tracking code

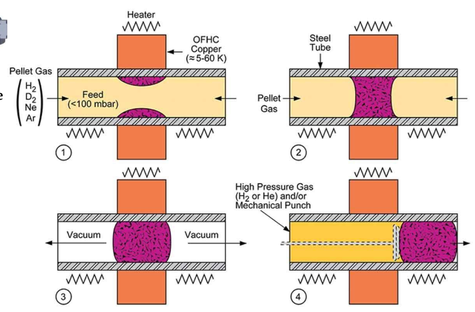
T.O.F. Speed (m/s)	Pixel Speed (m/s)	Vertical Dispersion (°)
152 ± 11	151 ± 2	-0.2 ± 0.4
57 ± 4	59 ± 1	-0.4 ± 0.2

Experimental Design

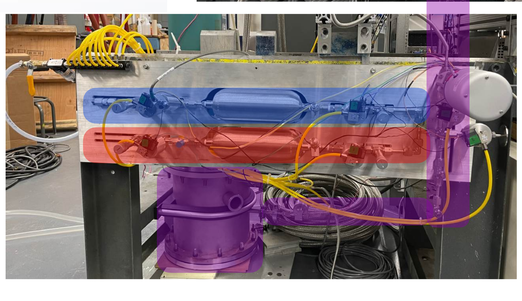








Pellet Formation [1]

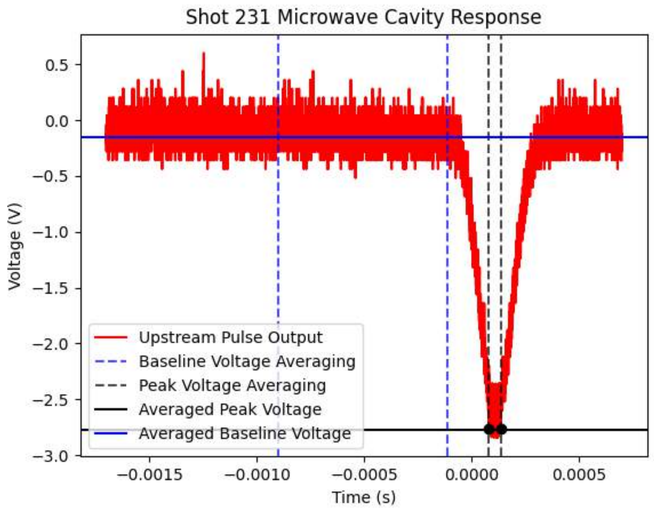


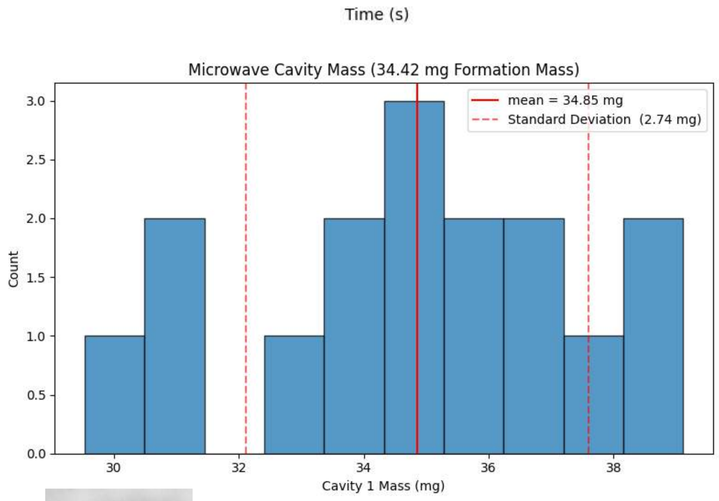
Gas Manifold

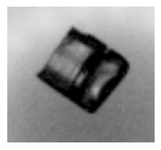
- Pellets frozen in injector barrel
- Multi-species gas manifold allows formation of homogenous and distinct shell + end-cap pellets
- 4 barrel heater locations allows formation flexibility
- Microwave cavities measure mass before and after passing through ablation zone
- Fast camera imaging (50k FPS) allows pellet tracking and secondary mass loss measurement
- Pellet dump mitigates post-injection pressure rise

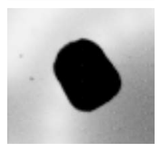
Multi-Species Mass

- Multi-species DMS pellet regimes lack density and fractional injection sublimation characterization
- Signal noise and detection of sublimated gas arrival in cavity caused 16% mass over measurement
- 16 34.42 mg H_2 formed pellets to calibrate MC mass code correction parameters
- 8 equal-volume Ne pellets used to approximate dielectric constant of Ne near triple point: ε = 1.25.
- 16 constant-volume H_2 +Ne pellets (0.5% - 15% Ne) have been fired
- Ne/H_2 dielectrics to be applied linearly by volume fraction investigating mixed pellet density and sublimation mass





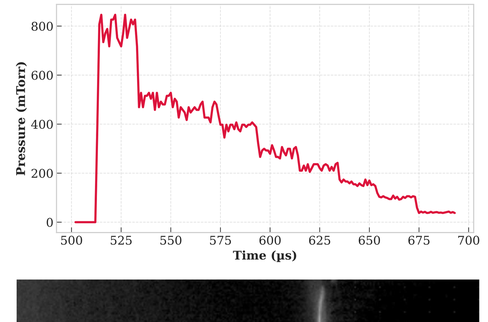


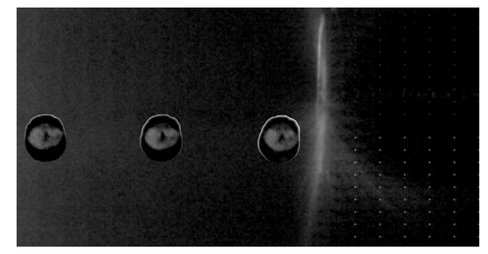


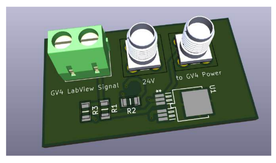
- Density will be examined near 2% Ne, where translucent to opaque transition is observed
- Inconsistent integrity above 15% Ne

Ablation Rate Measurement

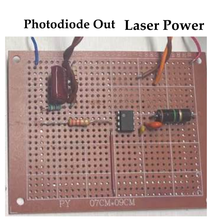
- E-beam requires minimization of post-injection pressure rise
- Guide 6° resin cone reduces sublimation in ablation chamber
- Active pumping on beam cathode
- Laser-photodiode trip wire installed for beam TTL triggering
- Pellet dump gate valve is pre-triggered by PCB with adjustable delay for predicted pellet speed to mitigate pressure back streaming
- With unoptimized pre-trigger delay pressure rise has been reduced from 800 mTorr to 90 mTorr
- E-beam has been installed, interlock and vacuum tests complete
- Beam 1-30kV scan this summer with H_2



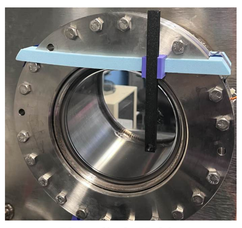




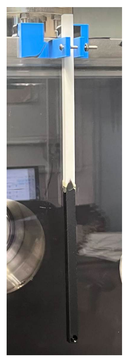
Dump Valve Trigger PCB



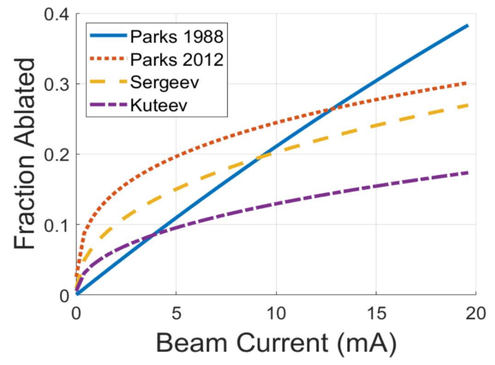
5V TTL Trigger Pulse



Photodiode Holder



Laser Holder



[2][3][4][5]

[1]: Combs, S. K., and L. R. Baylor. "Pellet-Injector Technology—Brief History and Key Developments in the Last 25 Years." Fusion Science and Technology, 73, no. 4, 2018, 493–518, DOI: 10.1080/15361055.2017.1421367.
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